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A magnetostratigraphic and secular variation study of Level Mountain, northern British Columbia

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Summary. Palaeomagnetic results are reported from 50 flows erupted by Level Mountain, which is a composite volcano in northern British Columbia. Fourteen magnetic polarity zones are represented, and these allow a magnetostratigraphy to be established which indicates that eruptive activity began about 6.5 Myr ago and continued up to very recent times with several hiatuses of up to 1 Myr or more. Most of the flows yield acceptable normal or reversed palaeopoles, but two of them record the Gauss/Matuyama polarity transition, and a further four yield a tight group of divergent directions, which are here interpreted as a geomagnetic excursion near the base of the Gauss Epoch. This observation lends support to an earlier suggestion (Bingham & Stone) that the geomagnetic field was abnormally disturbed about 3.5 Myr ago, at least locally in north-western North America.

The overall average palaeopole obtained after filtering out flows with high internal variance and those yielding divergent virtual geomagnetic poles (VGPs) lies at 256°E, 85°N ($N = 38$, $A_{95} = 7.2°$). A more representative palaeopole is obtained by combining the results with those of Souther & Symons from Mount Edziza, a closely related contemporaneous volcano some 100 km away. This pole lies at 292°E, 87°N ($N = 92$, $A_{95} = 4.1°$), and again includes the geographic pole within its 95 per cent confidence limits. However, the mean itself is not far-sided as Wilson's model predicts, neither is it left-handed as the analysis by Merrill & McElhinny predicts. The slight near-sidedness observed agrees with the global compilation of Wilson & McElhinny who show that the north-western region of North America is the only area, out of 15 regions worldwide, to yield a near-sided pole. The dispersion of the VGPs (angular standard deviation = 22°) is a little higher than would be expected from global summaries and secular variation models. This was also found to be the case in data from Alaska (Doell & Cox).

The angular dispersion, the occurrence of geomagnetic excursions, and the

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anomalous palaeopole offset, combine to suggest that a local long-lived source of geomagnetic perturbations may exist in the outermost core beneath this region.

Introduction

Level Mountain is a shield volcano covering some 1800 km^2 in northern British Columbia (131°20′W, 58°31′N). It consists of a volcanic plateau of average elevation about 1400 m, with a central region of peaks and ridges rising to a maximum elevation of about 2200 m.

Reconnaissance mapping (Gabrielse & Souther 1962) indicates a sequence of lavas of late Tertiary to Recent age, belonging to the alkalic series (Hamilton & Scarfe 1976). Several phases of extrusive activity can be discerned with mappable breaks between lava packages of distinct chemical and morphological features. Souther & Symons (1974) have described similar lavas and have investigated the palaeomagnetism of Mount Edziza, a closely related volcano some 100 km to the south-east. Souther (1970) has reviewed the regional geology and tectonics and concludes that late Cenozoic volcanism in this area may result from deep crustal rifting.

This paper describes the results of a palaeomagnetic study undertaken in parallel with a detailed petrological and geochemical investigation. The objectives were two-fold. The first was to provide a magnetostratigraphic profile of Level Mountain to aid in working out its detailed temporal evolution. The second was to obtain palaeomagnetic data pertinent to modern concepts of the morphology of the geomagnetic field and its secular variation.

Methods and results

Samples were collected with a hand-held diamond drill, cores being orientated by solar and/or topographic bearings. Two separate stratigraphic sections (Fig. 1) some 10 km apart were sampled, one representing the distal lower portion of the volcanic pile (the plateau section), the other representing the proximal upper portion (the stratocone section). Each section consists of 25 sampled flows. Field evidence suggests that there is no overlap between the sections, but that the gap involved is not large.

The stability of one specimen from each flow was tested by routine alternating field (AF) techniques, mostly in peak fields of 2.5, 5, 10, 20, 40 and 80 mT. The results were used to identify regions of directional stability, and thus to select appropriate treatments for the remaining specimens. As a check on directional stability all specimens were demagnetized at more than one field – generally two steps were sufficient, but in many cases three or more steps were carried out. The entire study involved about 1100 remanence measurements. Only one specimen from each core was measured as it has been demonstrated by Doell & Cox (1963, 1965) that very little additional accuracy is gained by multiple sampling.

The minimum dispersion criterion (Irving, Stott & Ward 1961) was used to identify the best estimate of the mean remanence direction at each site, and the results obtained are summarized in Table 1. At nine sites one or two specimens exhibited anomalous behaviour, either by systematically approaching the group formed by companion cores but never quite reaching that goal, or by stubbornly retaining a divergent direction. These specimens, which represent only 5 per cent of the data set ($N = 14$, parenthesized in Table 1), were excluded from further analysis. Although in part subjective, it is felt that this procedure yields the best available estimate of the true mean palaeomagnetic vector at each site.

The directional variance within each unit is generally small, with Fisher's precision parameter k greater than 100 in 34 out of 50 flows, and k less than 50 in only seven cases.

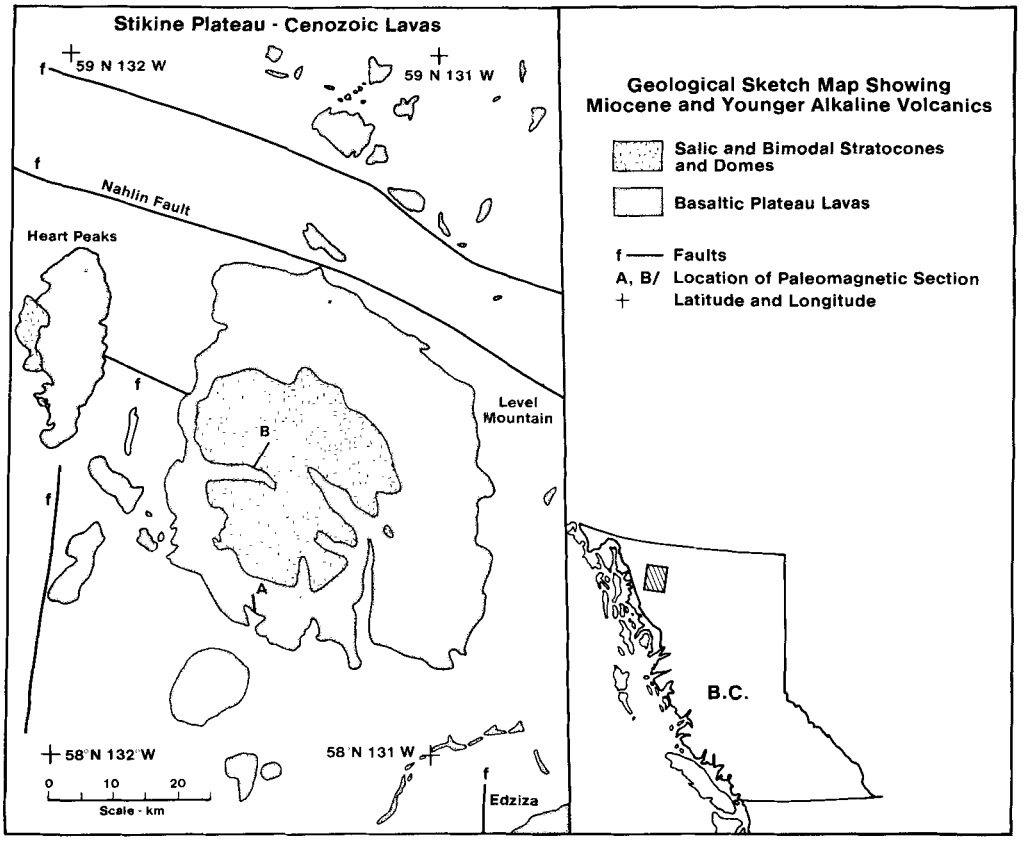


Figure 1. Geological sketch map showing location of sampling sections A and B.

Another convenient measure of the quality of the directional data is provided by the angular shift occurring at the demagnetization step used in the final analysis. When sites with poor internal grouping are excluded, as explained below, these angular shifts are generally quite small. Sixty-seven per cent of specimens undergo a shift of 3° or less, the corresponding values for $< 5°$ and $< 10°$ being 79 and 92 per cent respectively. Thus a small fraction of the specimens have no direct evidence of end-points, and their inclusion rests on agreement with cores from the same site, as reflected by the appropriate statistics given in Table 1.

Discussion

MAGNETOSTRATIGRAPHY

The polarity sequence is illustrated in Fig. 2 in terms of the latitude of site VGPs. The uppermost four flows (Bt–Bq) can be assigned unambiguously to the Brunhes Epoch since field evidence indicates that they post-date Quaternary continental glaciation. Beneath these four units there is a stratigraphic break, which is followed by five flows (Bp–Bl) with reversed polarity. These are assigned to the Matuyama Epoch. Eruptive activity seems to have been sparse and sporadic, and our sampling yields no evidence of the various 'events' in the Brunhes and Matuyama Epochs. With only nine flows representing some 2.5 Myr, and bearing in mind that the 'events' in question account for a total of only 0.3 Myr, this is

Table 1. Statistics for Level Mountain sampling sites.

Site	Elev	N	R	k	α_{95}	D	I	Lat	Long
Bt	7250	3(2)	2.9881	168	9.5	32.7	78.9	72.2	270.3
Bs	7200	3(2)	2.9932	294	7.2	45.5	55.1	52.7	336.7
Br	6625	5	4.9378	64	9.6	51.7	61.9	55.0	322.1
Bq	6580	3(2)	2.9886	175	9.3	30.4	54.0	59.0	355.9
Bp	6480	5	4.9656	116	7.1	216.0	-73.8	-71.6	119.0
Bo	6440	5	4.9387	65	9.5	184.0	-86.2	-65.5	51.3
Bn	6340	5	4.9983	2352	1.6	262.0	-78.3	-54.3	90.5
Bm	6160	5	4.9891	367	4.0	221.2	-62.3	-60.9	151.5
Bl	5835	5	4.9571	93	8.0	172.3	-65.6	-78.8	257.6
(By)	5780	5	4.7744	18	18.7	19.7	30 7	46.0	22.3
(Bk)	5745	5	4.7136	14	21.2	358.4	1.0	32.5	51.9
Bj	5690	5	4.9651	115	7.2	28.1	74.1	75.5	298.9
Bi	5675	5	4.9709	138	6.6	20.2	80.8	73.7	252.3
Bh	5655	5	4.9917	482	3.5	1.6	79.5	78.3	232.7
Bg	5580	5	4.9867	301	4.4	19.2	-1.2	(29.4	27.8)
Bf	5545	5	4.9621	106	7.5	12.4	5.1	(33.7	35.1)
Be	5475	5	4.9775	178	5.8	19.3	4.3	(32.1	27.0)
Bd	5460	6	5.9849	331	3.7	18.7	14.5	(37.3	26.4)
(Bc)	5450	5	4.8212	22	16.5	353.4	48.8	61.4	62.0
Bb	5430	5	4.9167	48	11.2	331.8	66.0	70.7	122.1
Ba	5420	4(1)	3.9968	937	3.0	4.3	53.9	66.3	41.2
(Bx)	5385	5	4.6887	13	22.2	197.5	-76.8	-79.2	93.1
Bu	5220	5	4.9351	62	9.8	348.9	72.6	84.1	144.0
Bv	5100	4(1)	3.9908	326	5.1	8.9	61.8	74.0	25.8
Bw	5000	5	4.9781	183	5.7	68.4	85.9	60.1	245.4
(Aa)	4560	5	3.9924	4	44.1	167.3	-64.0	-75.5	267.7
Ab	4505	5	4.9884	345	4.1	42.9	81.7	67.3	259.7
Ac	4465	5	4.9841	251	4.8	189.3	-67.3	-80.4	191.5
(Ad)	4440	2(2)	1.9869	76	29.0	169.5	-75.0	-83.5	0.4
Ae	4400	5	4.9978	1818	1.8	313.0	69.3	63.4	149.5
Af	4370	5	4.9773	176	5.8	294.2	81.7	61.0	198.2
Ag	4340	5	4.9894	377	3.9	280.4	76.9	54.0	185.1
Ah	4300	5	4.9886	351	4.1	15.8	66.7	77.2	356.4
Ai	4230	5	4.9940	667	3.0	63.0	86.2	60.7	243.9
Aj	4200	5	4.9878	328	4.2	310.1	80.0	66.0	191.3
AK	4170	5	4.9346	61	9.9	330.7	83.6	68.2	213.2
Al	4155	5	4.9804	204	5.4	308.6	70.1	61.7	154.6
Am	4135	5	4.9744	156	6.1	163.2	-45.6	-56.9	258.1
An	4125	5	4.9294	57	10.3	169.4	-46.2	-58.6	248.3
Ao	4115	4(1)	3.9986	2141	2.0	167.0	-75.0	-82.5	355.8
Ap	4100	5	4.9317	59	10.1	173.4	-84.4	-69.0	46.5
Aq	4065	5	4.9916	476	3.5	349.8	60.8	72.6	76.2
Ar	4030	4(1)	3.9957	698	3.5	44.7	57.8	55.3	334.5
As	3880	4(1)	3.9914	349	4.9	18.7	69.7	78.6	335.2
(At)	3860	4(1)	3.6531	9	33.1	156.6	-77.8	-76.2	8.3
Au	3815	5	4.9910	444	3.6	154.6	-60.7	-67.2	285.6
Av	3780	5	4.9259	54	10.5	178.8	-71.8	-88.5	256.5
Aw	3700	5	4.9698	133	6.7	0.0	21.3	(43.0	50.0)
Ax	3675	5	4.9955	889	2.6	147.5	-87.0	-62.9	42.9
Ay	3650	5	4.9770	174	5.8	65.5	-87.5	-55.7	41.9

Notes:

elev = elevation above sea-level (feet).
N = number of cores accepted (rejected).
R = vector resultant of N unit vectors.
k = Fisher's precision parameter.
α_{95} = half-angle of cone of 95 per cent confidence.
D = declination.
I = inclination.

perhaps not surprising. Indeed, it is possible that the gap between sites Bp and Bq could represent much of the 2.5 Myr involved.

Sites By and Bk yield mid-latitude VGPs and represent a transition to the next polarity epoch represented by the three sites Bj–Bh. Two fission-track dates on flows correlative with By yield ages of 2.46 ± 0.12 and 2.61 ± 0.12 Myr and we can therefore be fairly confident that this transition represents the Matuyama/Gauss boundary, which has an age of 2.48 Myr, according to a recent comprehensive analysis (Mankinen & Dalrymple 1979). We are indebted to Nancy Briggs and John Westgate of the University of Toronto for these fission-track dates, and the one quoted below.

Below site Bh there is another stratigraphic break and from here to the bottom of the volcanic pile it is no longer possible unambiguously to correlate the observed polarity pattern with the established time-scale. A fission-track age of 4.61 ± 0.26 Myr from a pitchstone correlated with site Bw suggests that the three basal flows in the B section (Bu–Bw)

most likely represent the Sidufjall event, younger and older boundaries of which are placed by Mankinen & Dalrymple (1979) at 4.32 and 4.47 Myr respectively. Above site Bu there is a break and the eight flows between it and the break below Bh can be tied to the polarity time-scale in several ways.

Sites Bg–Bd yield a tight group of low-latitude VGPs, and the five individual specimen results from site Bc are distributed between this group and the directions obtained from the underlying sites Bb and Ba (which explains the relatively low precision at Bc, see Table 1). These data suggest that the flows involved were erupted during a geomagnetic excursion or polarity transition. The possibility of a transition cannot be dismissed, but Bingham & Stone (1976) have reported excursions from Alaskan lavas 'considered to be about 3.4 Myr' old on the basis of K-Ar age determinations. Of three lava piles studied by Bingham & Stone (1976) two yielded divergent results interpreted as two separate field excursions. They concluded that the occurrence of excursions was higher during this period. We therefore place sites

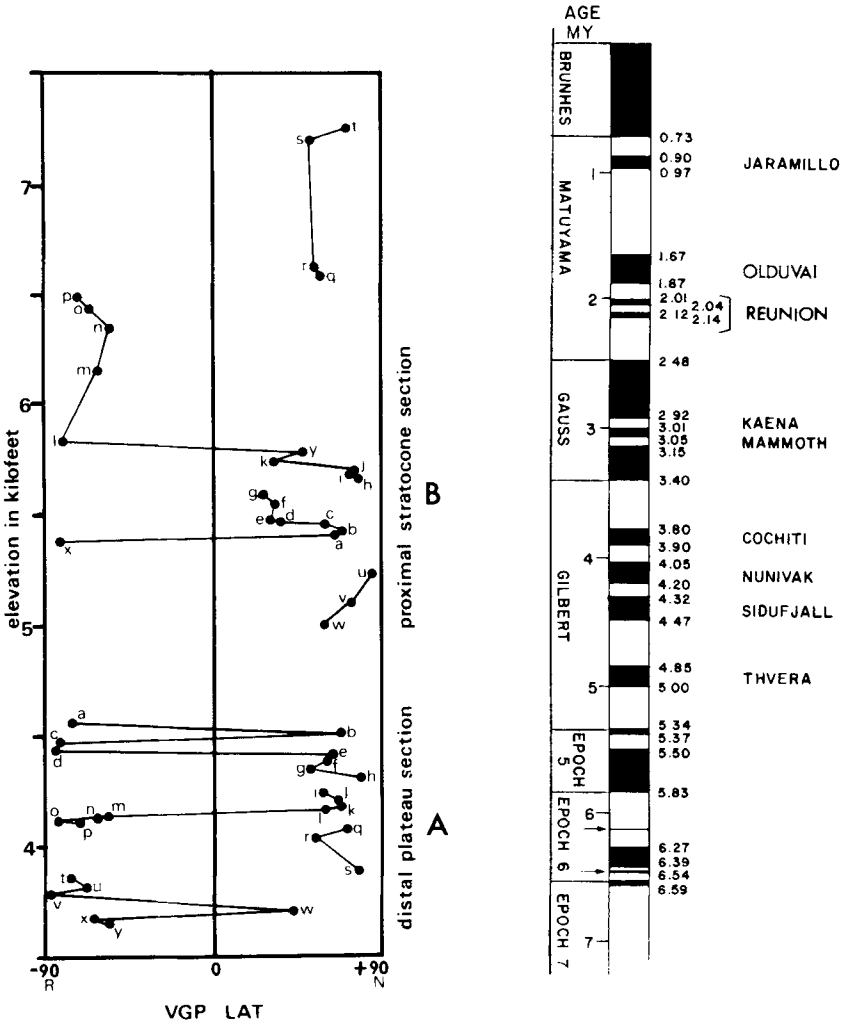


Figure 2. Latitude of site mean virtual geomagnetic poles. The time-scale at the right is taken from Mankinen & Dalrymple (1979) for ages less than 5 Myr, and from McDougall *et al.* (1977) for earlier times.

Bg–Ba in the lowermost normal polarity segment of the Gauss Epoch. Excursions of this kind need not be global in extent; they probably reflect rapid enhancement of localized magnetic sources in the uppermost outer core as modelled by Harrison & Ramirez (1975). The correlation adopted has the merit of placing sites Ba and Bb in the longest normal interval in this part of the standard time-scale, and of placing site Bx in the corresponding longest reversed interval, as would be expected on purely statistical grounds. If sites Bw to Bu do indeed represent the Sidufjall Event, then it appears that a considerable time interval (on the order of 0.5–1 Myr duration), including the Cochiti and Nunivak Events, is represented by the gap between sites Bu and Bx.

There are no radiometric data at all for the A section and thus any correlation with the standard time-scale must be regarded as rather tentative. Mankinen & Dalrymple (1979) do not carry their analysis beyond about 5 Myr, but the remarkably uniform extrusion rate occurring in the Borgarfjordur region of western Iceland has enabled McDougall *et al.* (1977) to extend the time-scale to about 6.5 Myr.

Taking the base of the B section as representing the Sidufjall event, and simply matching each lower magnetozone on a one-to-one basis with the sequence of normal and reversed intervals determined by McDougall *et al.* (1977) leads to the conclusion that the lowest sites (Ax and Ay) fall near the middle of Epoch 6 (about 6 Myr). This is a minimum estimate for the onset of volcanism in this area. If 'events' within the Gilbert Epoch or Epochs 5 and 6 have been missed by our sampling, which seems quite probable, the base of the section will be correspondingly older. In this context it is reasonable to suppose that the 8 flows Ae–Al represent the lower part of Epoch 5 (5.50–5.83 Myr), not the very short normal interval at the top (5.34–5.37 Myr) which consecutive matching implies. It also seems probable that the two extremely short events within Epoch 6 (indicated by small arrows in Fig. 2) would be missed at Level Mountain. If this is so then sites Aq–As may represent the normal interval 6.27–6.39 Myr, with sites Aw to Ay consequently falling in the upper part of Epoch 7 (about 6.6 Myr).

It is interesting to note that Souther & Symons (1974) who investigated the nearby, closely-related, Mount Edziza complex, concluded that their lowest flows were erupted about 6 Myr ago, which agrees very well with our interpretation of the Level Mountain data.

PALAEOPOLE

Some of the fifty site means summarized in Table 1 have relatively large internal variance and it is common practice to reject such sites as unreliable palaeomagnetic recorders, whatever the cause. Rather than select an arbitrary cutoff, we note that the present data fall naturally into two groups. Forty-three sites have α_{95} less than or equal to 11.2°, whilst five have α_{95} greater than 20°, the remaining two values being 16.5° and 18.7°. We therefore place a rejection threshold in the gap between 11.2° and 16.5°, with the result that seven sites are excluded. The appropriate site names are parenthesized in Table 1.

It is also normal practice to remove virtual geomagnetic poles (VGPs) lying in low latitudes, so as to avoid contamination from transition zones and/or geomagnetic excurions. Wilson, Dagley & McCormack (1972) reject VGPs with latitudes less than 50°, whereas Watkins (1973) uses a cut-off of 45° and Doell & Cox (1972) use 46°. The choice can be important in some cases, particularly when data sets are limited, but in the present case there are no poles between latitudes 45° and 50° so that choice is not critical. The five VGPs having latitudes less than 45° are parenthesized in Table 1.

Thirty-eight sites remain after filtering out sites with high internal scatter and low-latitude VGPs: 24 are 'normal' and 14 'reversed'. The two populations have similar scatter

and antiparallel mean directions (Table 2). The 'normal' pole is less than 3° from the inverted 'reversed' pole – certainly not a significant difference. It is therefore appropriate to combine the two populations as in Table 2. Note that more stringent rejection criteria for within-site scatter (α_{95} less than 10°, and less than 8°) cause no significant changes.

All the palaeopoles listed in Table 2 (i.e. normal, reversed, and the various combined) include the spin axis in their 95 per cent confidence circles, an observation which is relevant to modern concepts of the geomagnetic field. To a first approximation, the geomagnetic field, when averaged over a sufficient length of time, is that of a geocentric dipole. However, Wilson (1970, 1971, 1972) has pointed out that palaeomagnetic data for the last few million years imply that the expected field vector at any site is biased both in declination and inclination. Declinations are biased eastwards, and Wilson referred to the corresponding VGPs as 'right-handed'; inclinations are too shallow in the northern hemisphere, too steep in the southern, leading to 'far-sided' VGPs. Since the Level Mountain data include the spin axis within the circle of 95 per cent confidence no significant deviation can be claimed. However, the overall mean itself is right-handed – but it is not far-sided. Cox (1975) suggests a modification to Wilson's model in which far-sidedness is found only at low and mid-latitude sites, with near-sidedness prevailing at higher latitudes. Cox used the compilations of Creer, Georgi & Lowrie (1973) and Georgi (1974) to demonstrate that near-sidedness from high latitude sites appears to be a global phenomenon which he showed corresponds remarkably well to the morphology resulting from the 1965 IGRF values of g_1^0, g_2^0 and g_4^0. However, a more recent compilation (Merrill & McElhinny 1977) shows far-sided poles from all latitudes. Whatever global picture finally emerges, there does seem to be some form of regional anomaly. Poles from Alaska, western Canada and north-western USA are systematically near-sided. Of 15 worldwide regional poles given by Wilson & McElhinny (1974), that from this area was the only one to be significantly near-sided (see their fig. 2b – 16 VGPs give lat = 85.2°, long = 174.5°, $A_{95} = 4.40°$), and the present data from northern British Columbia support this observation (see Fig. 3).

Table 2. Summary of various mean values of palaeodirections and virtual geomagnetic poles for Level Mountain.

		N	R	k, K	α_{95}, A_{95}	D, long	I, lat	★
Normal	directions	24	23.2036	28.9	5.6°	008.4	+74.5	12°
	poles	24	21.9571	11.3	9.2°	261.2	+84.1	12°
Reversed	directions	14	13.4624	24.2	8.3°	180.0	−73.0	12°
	poles	14	12.7916	10.8	12.7°	061.2	−86.4	12°
Combined	directions	38	36.6558	27.5	4.5°	005.1	+74.0	12°
		34	32.8300	28.2	4.7°	008.2	+74.6	10°
		28	27.0198	27.5	5.3°	004.6	+74.5	8°
	poles	38	34.7391	11.4	7.2°	256.0	+85.0	12°
		34	31.0748	11.3	7.7°	261.6	+83.5	10°
		28	25.5326	10.9	8.6°	250.3	+84.5	8°

Notes:

N = number of sites.
R = resultant of N unit vectors.
k = Fisher's precision parameter for directions.
K = Fisher's precision parameter for VGPs.
α_{95} = radius of 95 per cent confidence cone for site directions.
A_{95} = radius of 95 per cent confidence cone for VGPs.
D, I = declination and inclination of mean directions.
long, lat = longitude and latitude of mean VGPs.
★ = within-site α_{95} cut-off (see text for further explanation).

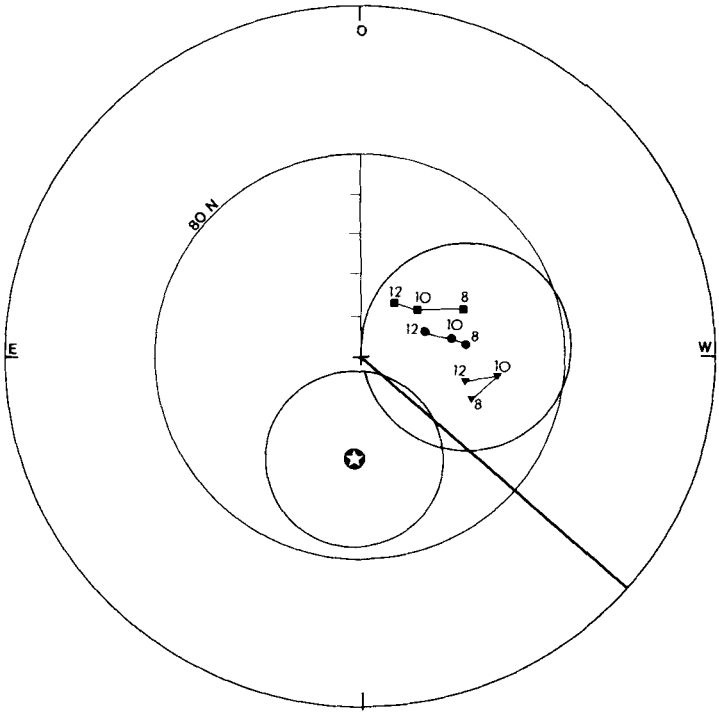


Figure 3. Equal-area plot of the polar cap region showing average virtual geomagnetic poles for Level Mountain (triangles) and Mount Edziza (squares). The circles represent the combined data set (see Tables 2 and 3). The numbers refer to the within-site α_{95} rejection criteria. The star represents the regional mean pole given by Wilson & McElhinny (1974) (see discussion in text). The radial line represents the longitude of Level Mountain.

The right-handed effect is more troublesome for two reasons – it is more difficult to establish observationally, and it is very difficult to explain theoretically. If it is valid, it implies a net toroidal field at the Earth's surface. Merrill & McElhinny (1977) argue that Wilson was misled by inadequate geographical distribution of sampling sites. They interpret their own comprehensive analysis in terms of a weak sinusoidal variation of declination as one proceeds along a given line of latitude, with easterly bias between 60°W and 120°E and westerly bias in the interval 120°E to 300°E. For the region in question here (i.e. northwestern North America) the earlier summary of Wilson & McElhinny (1974) agrees with this, but the Level Mountain data do not; they are biased slightly eastward (Table 2 and Fig. 3).

COMPARISON WITH MOUNT EDZIZA

Souther & Symons (1974) report palaeomagnetic data from Mount Edziza, a closely related volcano some 100 km to the south-east of Level Mountain. Extensive activity there covers the last 6 Myr, so the two volcanic centres are closely related in time as well as space. They obtain a mean pole less than 1° from the spin axis using all 84 flows sampled; but this includes low-latitude VGPs and sites with high internal scatter, of which there are a total of 30. No large changes occur when the rejection criteria used above are applied, but there is a noticeable tendency for the average pole to become systematically more near-sided and right-handed as the within-site α_{95} acceptance threshold is lowered from 12° to 10° and then to 8° (Table 3 and Fig. 3).

Table 3. Summary of palaeomagnetic data from Mount Edziza.

	N	R	k	α_{95}	D	I	★
Directions	54	52.7702	43.1	3.0°	006.1	+71.1	12°
	48	46.9331	44.1	3.1°	007.0	+71.8	10°
	36	35.2275	45.3	3.6°	010.2	+72.8	8°
	N	R	K	A_{95}	Long	Lat	★
Poles	54	50.8694	16.9	4.9°	329.9	+86.8	12°
	48	45.2069	16.8	5.2°	312.3	+86.5	10°
	36	33.8776	16.5	5.1°	297.8	+84.7	8°
	Combined with Level Mountain data from Table 2						
Poles	92	85.5245	14.1	4.1°	291.7	+86.8	12°
	82	76.2095	14.0	4.3°	283.6	+85.7	10°
	64	59.3679	13.6	5.0°	277.1	+85.0	8°

Notes:

(1) Mount Edziza data are taken from appendix III of Souther & Symons (1974). Reversed polarities have been inverted.
(2) Symbols have the same meaning as in Table 2.
(3) A VGP latitude cut-off of 50° was used for rejecting low-latitude VGPs (see text for further discussion).

In view of the close geological relationship between Level Mountain and Mount Edziza it is reasonable to combine the two data sets to obtain a better average palaeopole for northern British Columbia for the last few million years (Table 3). The combined data give no indication of the far-sidedness predicted by Wilson (1972) or the left-handedness predicted by Merrill & McElhinny (1977). Instead, the actual means listed in Table 3 are near-sided and right-handed regardless of the within-site α_{95} cut-off chosen. In the most conservative case (α_{95} less than 8°) the offset may just be significant (Fig. 3).

DISPERSION AND SECULAR VARIATION

Studies of this kind enable the secular variation as well as the position of the palaeopole to be studied. This is done in terms of the variance associated with the site VGPs. The best estimate of the total angular dispersion is given by

$$S_T = \left\{ \frac{1}{N-1} \Sigma D_i^2 \right\}^{1/2}$$

where N is the number of sites and D_i is the angle between the ith VGP and the mean. For the combined northern British Columbia dataset we obtain S_T of about 23°, the precise value depending on the within-site rejection criterion (see Table 4 for details). These values represent the total angular dispersion and require a correction for within-site dispersion (S_w) if overestimation of the dispersion of the actual geomagnetic field is to be avoided. In this case we obtain S_F = 21.9° with upper and lower 95 per cent limits 24.4° and 19.9° obtained from the tables given in Cox (1969). This value is a little higher than most theories of secular variation predict. In their analysis, McElhinny & Merrill (1975) obtain an average value of 17.9° with bounds of 19.0° and 16.9° for the global strip lying between latitudes 46° and 60° (based on 320 VGPs from four separate studies). The actual individual studies

Table 4. Dispersion statistics for combined Level Mountain and Mount Edziza VGPs.

N	S_T	S_F	S_U	S_L	*
92	22.5°	21.9°	24.4°	19.9°	12°
82	22.9°	22.3°	25.0°	20.1°	10°
64	23.8°	23.3°	26.5°	20.8°	8°

Notes:

S_T = total angular standard deviation.
S_F = angular standard deviation of palaeofield.
S_U, S_L = upper and lower 95 per cent confidence limits on S_F.
* = within-site α_{95} cut-off (see Tables 2 and 3).

involved yielded $14.2° \leqslant S_F \leqslant 22.3°$, and it is noteworthy that the highest of these is from Alaska. Perhaps these slightly high values are another manifestation of some kind of long-lived geomagnetic anomaly arising from a source in the uppermost core. Recall that this region yields 'anomalous' near-sided VGPs, and that at least for part of the time in question was apparently disturbed by an abnormally high rate of geomagnetic excursions.

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